Charge and the electric car

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain charge interactions and trace an electric car's energy transfers.

SUCCESS CRITERIA - I can...

1. Predict attraction or repulsion from charge signs.
2. Distinguish charge from energy.
3. Trace energy from a battery to moving wheels.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

An ideal blackbody has peak wavelength 966.7 nm. Use $\lambda_{max}\,T = 2.90 \times 10^{6}$ nm·K. Estimate temperature.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $E_{electrical} = E_{mechanical} + E_{thermal}$ + other transfers. Energy accounting for a motor with no net energy storage. A battery transfers energy; charge already exists in the wires.

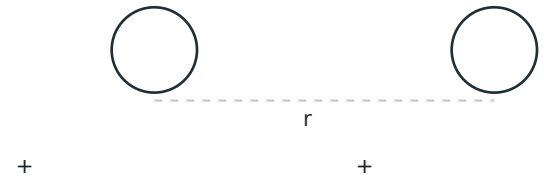
Worked reasoning: Rubbing can transfer electrons between materials. If one isolated object gains charge -3 units and another loses the same electrons, their charge changes add to zero.

Your turn: Draw before-and-after charge accounts for both objects. Distinguish transfer from creation of charge.

Marzano check - circle: ① ② ③

03 Represent and interpret

Draw force arrows for two positive charges, then for a positive and a negative charge.



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

04 Solve and justify

A battery transfers 120 J while the moving car gains 30 J kinetic energy. If it stays at the same height, how much energy enters other forms? Explain why charge is not used up.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

A student says the motor uses up the current so less returns to the battery. Explain the mistake for a steady series circuit.

Marzano check - circle: 1 2 3

06 Choose a method

Three instantaneous displacements are +4 mm, -6 mm and -8 mm. Find their sum and explain why one displacement value does not determine total wave energy.

Marzano check - circle: 1 2 3

07 Exit check

How can an initially neutral object be attracted to a charged object without having a net charge?

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

An initially neutral pair separates with charges +8 nC and -8 nC. One then shares all its net charge equally with an identical initially neutral conductor. Find the three final charges.

Marzano check - circle: 1 2 3